GGG-W-301b
January 8, 1963

SUPERSEDING
Fed. Spec. GGG-W-301a
May 27, 1954

FEDERAL SPECIFICATION

WHEEL, BUFFING AND POLISHING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers buffing sections, polishing wheels, and bobs made from a variety of fabrics, leathers, and other materials used for smoothing, coloring, shining, and cleaning of metallic and nonmetallic surfaces.

1.1.1. *Federal specification coverage.* This specification covers only the types and classes of buffing and polishing wheels generally purchased by the Federal government and is not intended to include all of the types and classes which are commercially available.

1.2 Classification.

1.2.1 *Types and classes.* Buffing sections, polishing wheels, and bobs covered by this specification shall be of the following types and classes as specified (see 6.1):

Buffing sections:

Type I.—Full-disc cloth.
  Class 1.—Loose.
  Class 2.—Sewed.
  Class 3.—Folded and sewed.
Type II.—Pieced-cloth.
Type III.—Biased-cloth.

Polishing wheels and bobs:

Type IV.—Canvas.
  Class 1.—Woven duck.
  Class 2.—Canvas belting.
Type V.—Felt.
Type VI.—Leather.
  Class 1.—Bullneck.
  Class 2.—Walrus.
  Class 3.—Sheepskin.
Type VII.—Cloth full-disc.
Type VIII.—Cloth pieced-disc.
Type IX.—Tampico brush.
  Class 1.—Untreated.
  Class 2.—Treated.

1.2.2 *Sizes.* Buffing sections, polishing wheels, and bobs covered by this specification shall be of the sizes listed in the applicable tables of section 3, and as specified (see 6.1).

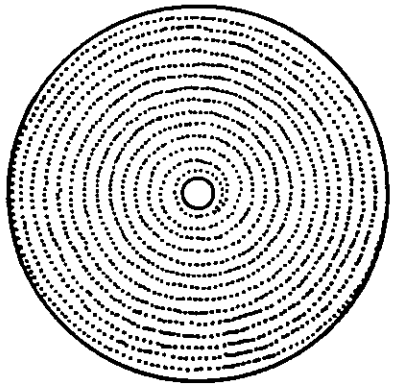
2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

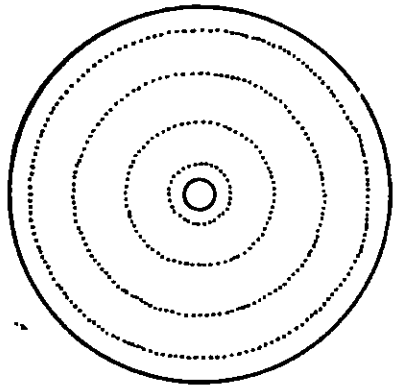
Federal Specifications:

KK-L-311—Leather, Methods of Sampling and Testing.
CCC-T-191—Textile Test Methods.
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585—Boxes, Wood, Wirebound.

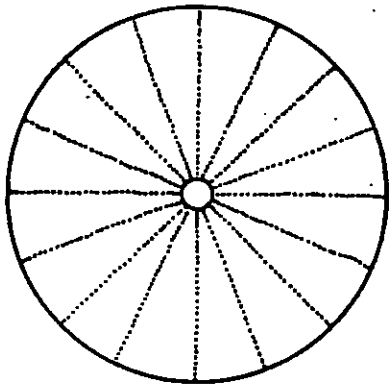
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.



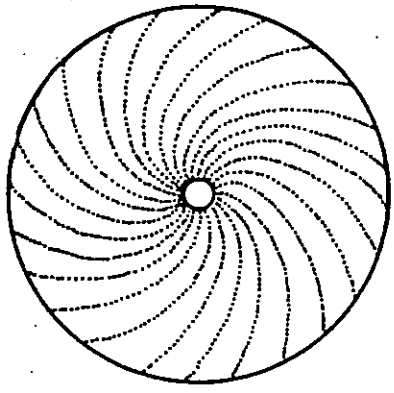
SPIRAL



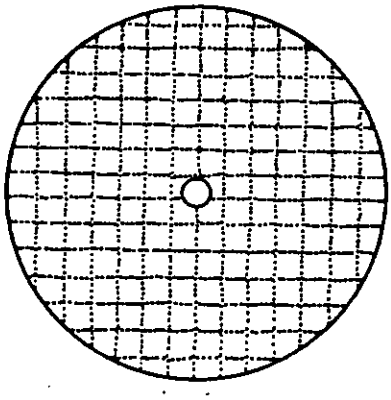
CONCENTRIC



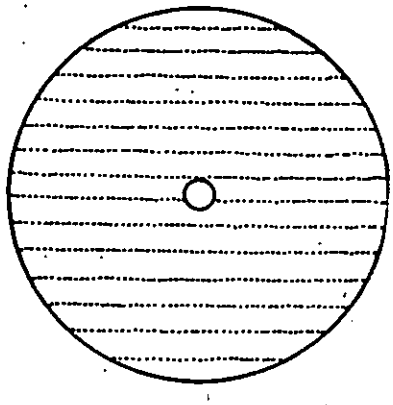
RADIAL



RADIAL ARC



SQUARE



PARALLEL

FIGURE 1.—Methods of sewing buffing wheels.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the current version before use.

PPP-B-591—Boxes, Fiberboard, Wood-Cleated.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-B-676—Boxes, Set-Up, Paperboard.

PPP-T-76—Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No 123—Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.

MIL-L-10547—Liners, Case, Waterproof.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of buff sections, polishing wheels, and bobs shall be new and shall be of the best quality and grade for the purpose intended.

3.2 Illustrations. The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of products otherwise conforming to the requirements of this specification.

3.3 Construction. Buffing sections, polishing wheels, and bobs requiring hand or machine sewing, glue, flexible cement, or combinations thereof, shall be constructed as specified (6.1).

3.3.1 *Sewing*. Figure 1 shows some of the typical methods of sewing buff sections. The procuring activity shall specify the type of sewing, spacing between rows, and number of rows required. Unless otherwise specified (6.1), machine sewing shall have from 5 to 6 stitches per inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

Table I. - Physical requirements, numbered duck and ounce duck

Number of duck, commercial designation	Weight, per square yard, minimum	Yard ply, minimum		Yarns per inch, minimum		Breaking strength, minimum	
		Warp	Filling	Warp	Filling	Warp	Filling
Hard texture	Ounces					Pounds	Pounds
12/0	48.00	9	9	20	12	800	430
8/0	40.00	8	14	26.5	12.5	800	665
2/0	31.90	5	7	24	16	465	435
1/0	30.31	5	6	24	16	450	405
1	28.71	5	5	26	19	440	370
2	27.12	5	4	26	19	420	345
3	25.53	4	5	29	20	390	330
4	23.93	4	4	31	22	375	300
5	22.33	3	4	35	23	345	285
6	20.74	3	3	35	25	335	250
8	17.55	3	3	43	25	285	210
10	14.35	3	3	45	27	245	160
11	12.75	3	2	45	34	240	140
12	11.16	2	2	45	35	195	120
Medium texture							
1	28.71	5	5	26	15	425	345
2	27.12	5	4	26	15	410	320
3	25.53	4	5	29	15	370	315
4	23.93	4	4	29	17	350	290
5	22.33	3	4	34	18	315	255
6	20.74	3	3	34	18	305	250
Ounce duck							
	8.25	2	2	60	58	125	120
	9.85	2	2	52	40	160	110
	12.29	3	2	44	34	210	130
	14.77	3	3	44	28	235	175
	15.90	3	3	45	34	245	200
	18.48	4	4	42	28	315	200

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:38Z
Check the source to verify that this is the current version before use.

3.3.2 *Trimming and balance.* Unless otherwise specified, buff sections shall be trimmed smooth on the face to the respective diameters. Buff sections, polishing wheels, and bobs shall be in balance within 1/16-inch ounce and the face shall be concentric with the arbor holes within 1/16 inch.

3.3.3 *Cloth.* Buff sections shall be constructed of flannel or muslin conforming to 3.3.3.1 and 3.3.3.2. The procuring activity shall specify the material required and whether bleached, unbleached, or colored cloth is required (6.1).

3.3.3.1 *Flannel.* Flannel used in buff sections shall be 8-ounce canton having a minimum thread count of 66/40 twill weave, fleeced nap, with a breaking strength not less than 80 pounds (warp) and 35 pounds (filling) Grab method.

3.3.3.2 *Muslin.* Unless otherwise specified, muslin used in buff sections shall have a weight of 2.6 yards per pound, thread count 86/93, plain weave, grey finish, with a breaking strength of not less than 90 pounds (warp), 65 pounds (filling) Grab method.

3.3.4 *Canvas.* Buff sections of canvas shall be constructed of duck or belting as specified in 3.3.4.1 and 3.3.4.2 (6.1).

3.3.4.1. *Duck.* Duck used in buff sections shall be single ply in accordance with table I for the thickness selected by the contractor

3.3.4.2. *Belting.* Belting used in the buff sections shall comply with table II, ply to be selected by the contractor to meet section thickness requirements as desired.

3.3.5 *Felt.* Wool felt shall consist of wool and cotton. The felt shall be substantially free from "wool fat" and "wool perspiration", and shall contain no animal fiber other than wool. The wool felt shall be in accordance with table III. The degree of hardness for the various grades of felt such as rock hard, hard, medium, and soft shall be to the tolerances specified in table IV and as specified (6.1).

Table II. - Belting

	3 Ply	4 Ply	6 Ply
Thread count-Warp (per inch, min.)	8/4-96	8/4-128	8/4-192.
Thread count-Binder	8/4-16	8/4-16	8/4-16.
Thread count-Filling	8/6-36	8/6-48	8/6-72.
Ply	3	4	6
Weight	40 lbs. per-100 sq. ft.	64 lbs. per-100 sq. ft.	92 lbs. per-100 sq. ft.
Weave	Conventional	Conventional	Conventional
Breaking strength, lbs. min.	800	1,000	1,400.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the current version before use.

Table III. - Content and properties of wool felt

Actual wool content (minimum)	Soluble matter (maximum)			Ash content
	In CCl_4	In water	In CCl_4 and water	
Percent 95	Percent 2.5	Percent 2.5	Percent 3.0	Percent 2.5

3.3.6 *Leather.* Leather shall be bullneck, sheepskin, or walrus as specified (6.1). The leather shall be free from loose fleshy spots or other blemishes, and shall conform to table V.

3.3.7 *Hubs.* Hubs or inserts shall be furnished as an integral part of the buff and shall be fabricated from plaster, leather, metal, or cloth as specified (6.1).

3.4 Type I, full disc cloth.

3.4.1 *Material.* Material shall be muslin or flannel as specified in 3.3.3. In addition the sections shall be full disc each ply.

3.4.2 *Construction.*

3.4.2.1 *Class 1, loose.* The loose full-cloth disc shall not be sewed or shall be sewed once concentric with the arbor hole as specified in 3.3.1.

3.4.2.2. *Class 2, sewed.* The sewed full-cloth disc shall be made from full circles of material with every two layers of cloth rotated slightly to produce a more uniform wearing surface on the face of the buffing section. Sewing shall be as specified in 3.3.1.

3.4.2.3 *Class 3, folded and sewed.* The folded and sewed full-cloth disc shall be faced with two cover discs not folded. The balance of the discs shall be folded into a wedge shape and arranged with points inward, each disc interleaved with the next. The buffing section shall be sewed together by sewing two rows 1/4 inch apart. The location of the outer row shall be at one-half of the buff diameter. Both rows shall be concentric with the arbor. Starting at a point one-third of the buff diameter, sew spirally in rows 1/4 inch apart. Continue sewing spirally to the arbor hole (3.3.1). Folding shall be single, double, or triple as specified (6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

Table IV. - *Degree of hardness for various grades of felt*

Nominal Thickness, inches	Soft					Medium				
	Weight, lbs/sq. yd.			Thickness, inches		Weight, lbs/sq. yd.			Thickness, inches	
	Nominal	Limits		Limits		Nominal	Limits		Limits	
		Min.	Max.	Min.	Max.		Min.	Max.	Min.	Max.
1 ----	16.00	15.20	16.80	0.964	1.086	20.00	19.00	21.00	0.970	1.030
	Hard					Rock Hard				
	Weight, lbs/sq. yd.			Thickness, inches		Weight, lbs/sq. yd.			Thickness, inches	
	Nominal	Limits		Limits		Nominal	Limits		Limits	
		Min.	Max.	Min.	Max.		Min.	Max.	Min.	Max.
1 -----	28.00	24.60	27.40	0.975	1.025	32.00	30.40	33.60	0.981	1.019

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

Table V. - Leathers

	Bullneck	Walrus	Sheepskin
Grease content, percent	2-4	---	---
Type of tanning	Rough oak(vegetable)	Not required	Vegetable.
PH factor	2.5-5.0	---	3.1-3.6
Tensile strength, p.s.i.	2,000	---	2,127

3.5 Type II, pieced-cloth.

3.5.1 *Material.* Material shall be muslin as specified in 3.3.3.2.

3.5.2 *Construction.* Pieced-cloth buffing sections shall be made of flat strips of material between full disc covers. No strip of material shall be used which is less in width than one-third of the diameter of the finished buff. The strips shall be laid side by side, in layers and the direction of each succeeding layer or ply shall be at various angles to the adjoining layers. No pieces other than strips shall be used in the construction of a pieced-cloth buffing section. Sewing shall be as specified in 3.3.1.

3.5.3 *Detail requirements.* Detail requirements for type II, buffing sections shall be as specified in 3.3.3.

3.6 Type III, biased-cloth.

3.6.1 *Material.* Material shall be flannel or muslin as specified in 3.3.3.

3.6.2 *Construction.* The biased-cloth buffing section shall be made of single strips of biased-cut material, so that they will not ravel or whirl apart, securely held along the center by concentric sewing to a hub in which the arbor hole has been punched or drilled (figure 2).

3.6.3 *Detail requirements.* Detail requirements for type III buffing sections shall be as specified in 3.3.3.

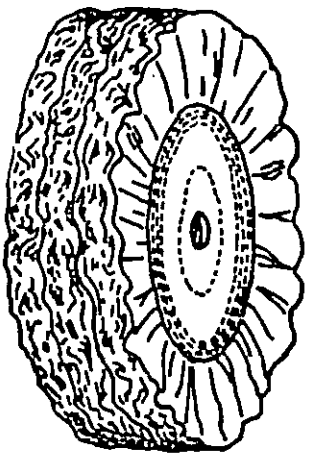


FIGURE 2.—Biased cloth buff.

3.7 Type IV, canvas.

3.7.1 *Material.* Material shall be woven cotton duck or canvas belting as specified in 3.3.4.

3.7.2 *Construction.*

3.7.2.1 *Class 1, woven duck.* The canvas-woven-duck polishing wheels shall be of the full-disc type. Several layers or canvas discs shall be sewed together and then a sufficient number of sewed sections shall be glued together to form the required face width. Sewing of discs into sections shall be in accordance with 3.3.1.

3.7.2.2. *Class 2, canvas belting.* The canvas belting polishing wheels shall be of the full-disc type with sufficient discs glued together to form the required face width.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

Table VI. - Requirements for polishing wheels

Wheel diameter, inches	Face width, inches					Arbor-hole diameter, inches						
	1/2	1	1-1/2	2	3	1/2	5/8	3/4	7/8	1	1-1/4	1-1/2
6	x	x	x	x	x	x	x	x	x	x	x	x
8	x	x	x	x	x	x	x	x	x	x	x	x
10	x	x	x	x	x	x	x	x	x	x	x	x
12	x	x	x	x	x	x	x	x	x	x	x	x
14	x	x	x	x	x	x	x	x	x	x	x	x
16	x	x	x	x	x	x	x	x	x	x	x	x

3.7.3 *Detail requirements.* Type IV polishing wheels shall conform to the requirements of table VI. The density or degree of hardness of the type IV, class 1 wheels shall be hard, medium hard, medium, medium soft or soft as specified. (6.1) Type IV, class 2 shall be medium density only.

3.8 Type V, felt.

3.8.1 *Material.* Material shall be felt as specified in 3.3.5.

3.8.2 *Detail requirement.* Type V polishing wheels shall be of the square-face type and shall conform to the requirements of table VI for wheels or table VII for bobs.

3.9 Type VI, leather.

3.9.1 *Material.* Material shall be bullneck, sheepskin, or walrus as specified in 3.3.6.

3.9.2 *Construction.*

3.9.2.1 *Class 1, bullneck.* Bullneck wheels and bobs shall be full disc made by gluing together sufficient layers of leather to produce a wheel of desired face width.

3.9.2.2. *Class 2, walrus.* Walrus wheels and bobs shall be full-disc and of single section.

3.9.2.3 *Class 3, sheepskin.* Sheepskin wheels shall be full-disc type. They shall be loose, sewed once concentric with the arbor hole, sewed sections glued together, or quilted types. Construction shall be as specified (6.1).

3.9.3 *Detail requirements.* Type VI polishing wheels shall be of the square-face type and shall conform to the requirements of table VI for wheels or table VII for bobs.

3.10 Type VII, cloth full-disc.

3.10.1 *Material.* Cloth full-disc polishing wheels shall be constructed of buff sections as specified in 3.4.

3.10.2 *Construction.* The sections shall be joined together by flexible cement, glue, or hand sewing glued around the arbor hole, as specified (6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

Table VII. - Requirements for polishing bobs

Wheel diameter, inches	Face width, inches					Arbor-hole diameter, inches
	1/8	1/4	3/8	1/2	3/4	1/8
1	x	x	-	-	-	x
1-1/4	-	x	x	-	-	x
1-1/2	-	x	x	x	-	x
2	-	x	x	x	-	x
2-1/2	-	-	x	x	-	x
3	-	-	x	x	x	x
4	-	-	x	x	x	x

3.10.3 *Detail requirements.* Type VII polishing wheels shall conform to the requirements of table VI. The density of the wheel shall be hard, medium or soft as specified (6.1).

3.11 Type VIII, cloth pieced-disc.

3.11.1 *Material.* Cloth pieced-disc polishing wheels shall be constructed of buff sections as specified in 3.5.

3.11.2 *Construction.* Construction shall be as specified in 3.10.2.

3.11.3 *Detail requirements.* Type VIII requirements shall be as specified in 3.10.3.

3.12 Type IX, tampico brush.

3.12.1 *Material.*

3.12.1.1 *Class 1, untreated.* The class 1 wheels shall be made from untreated white or gray tampico fiber.

3.12.1.2 *Class 2, treated.* The class 2 wheels shall be made of the same material as specified in 3.12.1.1 except that they shall be treated after assembly. The fiber shall be uniformly coated and the space between the bristles filled with a semisolid, adhesive, homogenous mixture of suitable gums or resins of oils, which will prevent undue splitting or wearing away of the bristles during use. and permit the fiber to carry a large amount of cutting compound.

3.12.2 *Construction.* Type IX tampico-brush polishing wheels shall be made with either a metal center and flanges, cardboard-fiberboard or with a wooden hub, as specified. The wheels shall be constructed in such a manner as to provide a uniformly dense face with suitable means for securely holding the fiber in place. When wooden hubs are specified, they shall be close-grained, thoroughly seasoned hardwood, free from knots and other injurious defects. Wooden hubs shall have the fiber secured in the holes by the copper wire-drawn method, and shall have a face width of approximately 1/4 inch per row of holes (6.1).

3.12.3 *Detail requirements.* Type IX polishing wheels shall conform to the requirements of table VI for wheel diameter, hole size, and face width specified (6.1).

3.13 Identification marking.

3.13.1 *Civil agencies.* Each buff section, polishing wheel, and large bob shall be marked in a plain and permanent manner with the name of the manufacturer or trademark of such known character that the source of manufacture can be readily identified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

3.13.2 *Military agencies.* Unless otherwise specified (6.1), the following applicable information shall be permanently and legibly marked on each buff section, polishing wheel, and large bob:

Type
Class
Size
Fed. Spec. No. GGG-W-301
Mfr's. Model No.
Mfr's Serial No.
Order No.
Stock No.
Mfr's. Name or trademark

Where the size of small bobs prohibits the above requirements, marking shall be on the unit package only in accordance with 5.3.

3.14 Workmanship. Buffing sections, polishing wheels, and bobs shall be new and of the quality of workmanship necessary to produce items of good appearance, having all operating characteristics found in first-class commercial quality items of a similar nature.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 *Lot.* All wheels of the same type, class, and size offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.2.2 *Sampling for visual and dimensional examination.* Wheels shall be selected at random from the lot in accordance with MIL-STD-105 at Inspection Level II for the examination specified in 4.3. The Acceptable Quality Level shall be 1.5 percent defective.

4.2.3 *Sampling for weighing test.* The same wheels employed in 4.2.2 shall be used for the test of 4.4.1. The Acceptable Quality Level shall be 2.5 percent defective.

4.2.4 *Sampling for tests of materials.* Sample pieces of cloth, duck, canvas, and leather shall be selected in accordance with MIL-STD-105 at Inspection Level L-3 for testing in accordance with the applicable portions of CCC-T-191 and KK-L-311 for textiles and leathers, respectively. If any specimen fails in any test, the lot represented shall be rejected.

4.3 Visual and dimensional examination. The wheels shall be examined and measured to verify conformance to all requirements of this specification which do not involve tests. Examination shall be conducted as specified in table VIII. Any wheel having one or more defects shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:38Z
Check the source to verify that this is the current version before use.

Table VIII. - Classification of defects in accordance with Standard MIL-STD-105

Catagories	Defects
Critical:	
1	None defined.
Major:	
101	Type, class, diameter, face width, and arbor hole diameter not as specified (types I, II, and III only).
102	Material defective, not new, or not as specified (types I through IX).
103	Sewing (if applicable) defective, incomplete, number of stitches per inch not as required, or type of sewing not as specified (types I through IX).
104	Buff section not trimmed smooth (types I through IX).
105	Cloth not full disc each ply (type I only), layers not rotated slightly to produce even wear (type I, class 2 only), or folding (type I, class 3 only) not as specified.
106	Width of strips less than the specified minimum, or direction of succeeding layers not at various angles to adjoining layers (type II only).
107	Strips not biased-cut or strips not securely sewn to hub (type III only).
108	Type, class, diameter, face width, and density and arbor hole diameter not as specified (types IV through IX only)
109	Canvas layers not full-disc type, not sewn together, or sewed sections not adequately glued (type IV, class 1 only).
110	Canvas layers not full-dics type or layers not adequately glued (type IV, class 2 only).
111	Felt not substantially free from wool fat and wool prespiration (type V only).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

Table VIII. - Classification of defects in accordance with Standard MIL-STD-105 (Continued)

Catagories	Defects
Major:	
112	Leather not free from loose fleshy spots and other blemishes, layers not full-disc type, layers not adequately glued (if applicable), or type of construction not as specified (type VI only).
113	Cloth not full disc each ply, layers not rotated slightly to produce even wear, or construction not as specified (type VII only).
114	Width of strips less than the specified minimum direction of succeeding layers not at various angles to adjoining layers, or type of construction not as specified (type VIII only).
115	Face of wheel not uniformly dense, fibers not securely held in place, wooden hub (if applicable) not close grained, thoroughly seasoned hardwood, or wooden hub not free from knots and cracks (type IX only).
116	Fibers not uniformly coated, space between bristles not filled, or coating not as specified (type IX, class 2 only).
Minor:	
201	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent (types I through IX).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

4.4 Test procedures.

4.4.1 *Weighing.* Each of the sample wheels shall be weighed separately and compared with all of the wheels. Any wheel which deviates in unusual degree from the other wheels shall be rejected.

4.4.2 *Possible test failures are defined as follows:*

(a) Buff section not in balance within the specified maximum value (types I, II, and III only).

(b) Wheels not in balance within the specified maximum value (types IV through IX only).

(c) Face of buff section not concentric with the arbor hole within the specified value (all types).

4.5 Inspection of preparation for delivery. Sample wheels, packages, and packs shall be selected and inspected in accordance with MIL-P-116 to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

(Civil agencies. Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of preservation, packaging, and packing.)

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (6.1).

5.1.1 *Level A.* Wheels shall be individually packaged and unit protected in accordance with method IC of MIL-P-116. Unless otherwise specified (see 6.1), the submethod selection shall be at the option of the contractor. Wheels requiring moth protection shall be provided with naphthalene flakes or paradichlorobenzene crystals. Unit containers shall conform to PPP-B-566, PPP-B-676, or PPP-B-636 at the option of the contractor. Box closure shall be in accordance with the applicable box specification or appendix thereto.

5.1.2 *Level C.* Wheels shall be afforded preservation and packaging in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 *Level A.* Wheels, packaged as specified (see 5.1), shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	*Type or class*
PPP-B-576	Class 2
PPP-B-585	Class 3 use
PPP-B-591	Overseas type
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class 2

5.2.1.1 Shipping containers shall have case liners conforming to MIL-L-10547. Case liners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Case liners for class 2 fiberboard boxes conforming to PPP-B-636 may be omitted provided all corners and edge seams and manufacturer's joints are sealed with minimum 1-1/2-inch wide tape conforming to PPP-T-76. Boxes shall be closed, strapped, or banded in accordance with the applicable box specification or appendix thereto. The gross weight of wood-cleated boxes shall not exceed 200 pounds.

5.2.2 *Level B.* Wheels, packaged as specified (see 5.1), shall be packed in boxes conforming to any of the following specifications at the option of the contractor:

Specification	*Type or class*
PPP-B-576	Class 1
PPP-B-585	Class 1 or 2 use
PPP-B-591	Domestic type
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Class 2

5.2.2.1 Box closures shall be as specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.3 *Level C.* Wheels, packaged as specified (see 5.1) shall be packed in containers of the type, size and kind commonly used for the purpose, in a manner that will insure safe delivery and acceptance at destination. Shipping containers shall comply with rules and regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military agencies.* In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.2 *Civil agencies.* Shipment markings shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Material. (3.3.3, 3.3.3.2, 3.3.4, 3.3.5, 3.3.6, and 3.3.7).
(*c*) Size—Outside diameter, thickness, and arbor hole (tables VI and VII).
(*d*) Type and class (1.2).
(e) Sewing—type, distance between number of rows, stitches per inch (3.3.1, and figure 1).
(*f*) Construction. (3.3, 3.9.2.3, 3.10.2, and 3.12.2).
(*g*) Density (3.3.5, 3.7.3, and 3.10.3)
(*h*) Number of folds (3.4.2.3)
(*i*) Level of preservation and packaging, and level of packing required (5.1 and 5.2).

6.1.1 *Data with bid.* Each bidder shall be required to furnish with his bid complete descriptive data covering the items he proposes to furnish under this specification and the invitation for bids.

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail: Wheels, buffing and polishing, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor: Wheels, buffing, and polishing not otherwise indexed.
Motor volume minimum weight 30,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—Ord
Navy—Ships
Air Force—WRA

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the current version before use.